

$$\frac{1}{2}m\vec{v}^2 = \frac{1}{2}m\vec{v}_0^2 + \vec{F}\cdot\vec{r} - \vec{F}\cdot\vec{r}_0$$

$$\vec{a}\vec{F}$$

$$m??$$

$$\vec{F} = m\vec{a} \tag{1}$$

$$?? \quad \vec{F} = 0\vec{a} = 0$$

$$\vec{F}[kgm/s^2]$$

$$1N = 1kgm/s^2 \tag{2}$$

$$\spadesuit \quad 1\spadesuit$$

$$xmFF_0$$

$$x(t_0)=x_0 \tag{3}$$

$$\dot{x}(t_0)=v_0 \tag{4}$$

$$\# \quad \#$$

$$F_0 = m\ddot{x} \tag{5}$$

$$m$$

$$x = \frac{F_0}{2m}t^2 + At + B \tag{6}$$

$$x(t_0) = \frac{F_0}{2m}t_0^2 + At_0 + B = x_0 \tag{7}$$

$$\dot{x}(t_0) = \frac{F_0}{m}t_0 + A = v_0 \tag{8}$$

$$AB$$

$$A=v_0 - \frac{F_0}{m}t_0 \tag{9}$$

$$B=\frac{F_0}{2m}t_0^2 - v_0t_0 + x_0 \tag{10}$$

$$x = \frac{F_0}{2m}(t-t_0)^2 + v_0(t-t_0) + x_0 \tag{11}$$

♠ 2♠
 xmF

$$F=\left\{\begin{array}{ll}0 & (t<0) \\ F_0(0\leq t\leq t_0) & \\ 0 & (t_0<t)\end{array}\right. \tag{12}$$

$$F_00\leq t\leq t_0t<0x=x_0$$

♠ ♠
 $??t<0x=x_00\leq 0\leq t_0$

$$x=\frac{F_0}{2m}t^2+At+B \tag{13}$$

$$A,Bt_0< t$$

$$x=Ct+D \tag{14}$$

$$t=0$$

$$x(0){=}B=0 \tag{15}$$

$$\dot{x}(0){=}A=0 \tag{16}$$

$$0\leq t\leq t_0$$

$$x(t)=\frac{F_0}{2m}t^2 \tag{17}$$

$$t=t_0$$

$$x(t_0){=}\frac{F_0}{2m}t_0^2=Ct_0+D \tag{18}$$

$$\dot{x}(t_0){=}\frac{F_0}{m}t_0=C \tag{19}$$

$$C{=}\frac{F_0}{m}t_0 \tag{20}$$

$$D{=}{-}\frac{F_0}{2m}t_0^2 \tag{21}$$

$$x(t)=\left\{\begin{array}{ll}0 & (t<0) \\ \frac{F_0}{2m}t^2 & (0\leq t\leq t_0) \\ \frac{F_0}{2m}t_0(t-t_0) & (t_0< t)\end{array}\right. \tag{22}$$

$$0\leq t\leq t_0t_0< t$$

♠ 3♠
 xym

$$x(t)=a\cos\theta(t) \tag{23}$$

$$y(t)=a\sin\theta(t) \tag{24}$$

$$\sharp\,\sharp$$

$$\ddot{x}\!=\!-a\left(\ddot{\theta}\sin\theta+\dot{\theta}^2\cos\theta\right)\tag{25}$$

$$\ddot{y}\!=\!a\left(\ddot{\theta}\cos\theta-\dot{\theta}^2\sin\theta\right)\tag{26}$$

$$xF_xyF_y$$

$$F_x\!=\!-\frac{a}{m}\left(\ddot{\theta}\sin\theta+\dot{\theta}^2\cos\theta\right)\tag{27}$$

$$F_y\!=\!\frac{a}{m}\left(\ddot{\theta}\cos\theta-\dot{\theta}^2\sin\theta\right)\tag{28}$$

$$\theta=\omega t$$

$$F_x\!=\!-\frac{a}{m}\omega^2\cos\omega t\tag{29}$$

$$F_y\!=\!\frac{a}{m}\omega^2\sin\omega t\tag{30}$$

$$\tag{31}$$

$$\vec{F}_{12}\vec{F}_{21}=-\vec{F}_{12}$$

$$m_1m_2\vec{x}_1(t)\vec{x}_2(t)\vec{F}_{12}\vec{F}_{21}$$

$$\vec{F}_{12}\!=\!m_1\ddot{\vec{x}}_1\tag{32}$$

$$\vec{F}_{21}\!=\!m_2\ddot{\vec{x}}_2\tag{33}$$

$$\vec{F}_{12}=-\vec{F}_{21}$$

$$m_1\ddot{\vec{x}}_1+m_2\ddot{\vec{x}}_2=0\tag{34}$$

$$m_1\ddot{\vec{x}}_1+m_2\ddot{\vec{x}}_2$$

$$\spadesuit\;4\spadesuit\\ \theta m??N_1N_2$$

$$\sharp\,\sharp\\ mg$$

